Is anaemia at 34 weeks of pregnancy linked to higher anxiety and lower well-being in pregnant women in Jaipur

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Introduction

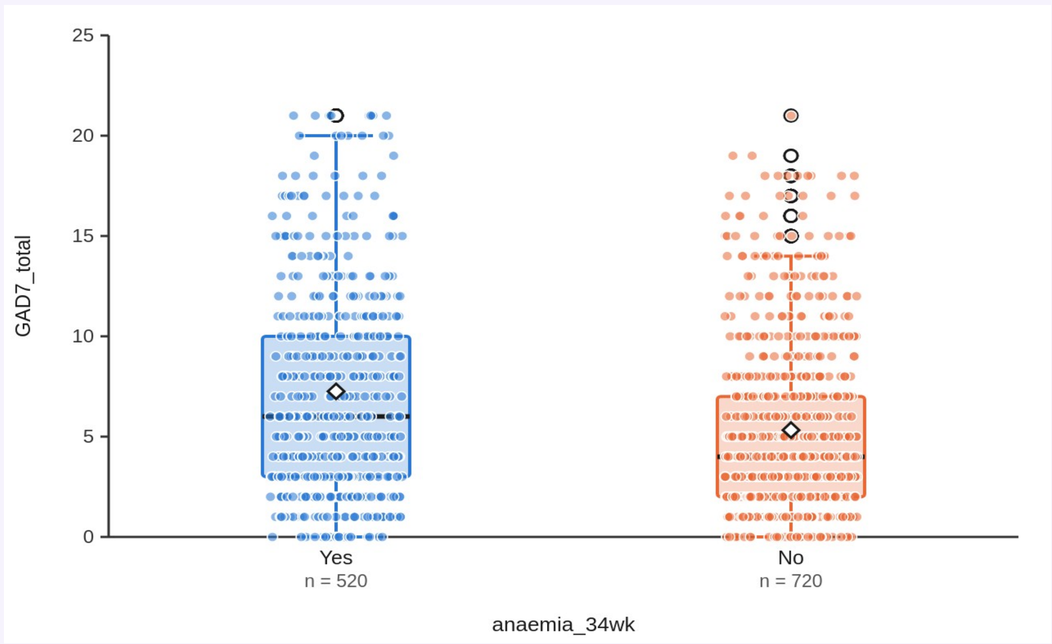
- Anaemia is one of the most common health problems of pregnancy, and the burden is highest in South Asia (Kassebaum et al., 2014).
- Anxiety is also common in pregnancy.
- A search of OpenAlex and PubMed for 2010 to 2025 found studies of anaemia and of anxiety in pregnancy, but none in Rajasthan that measured haemoglobin and validated anxiety, self-esteem and…
- The present study asked: Is anaemia at 34 weeks of pregnancy linked to higher anxiety and lower well-being in pregnant women in Jaipur?
- H1: Women with anaemia at 34 weeks will have higher GAD-7 scores than women without anaemia.
- H2: Higher self-esteem will go with higher life satisfaction, and higher anxiety with lower life satisfaction.
- H3: Haemoglobin will rise between 16 and 34 weeks with iron-folic acid tablets.

Methods

- Design. This was a cross-sectional survey.
- Participants. Pregnant women aged 18 or over enrolled at 12 to 16 weeks at two government hospitals in Jaipur (the simulated Janani cohort).
- Measures. Anxiety. Generalized Anxiety Disorder 7-item scale (GAD-7, 7 items) (Spitzer et al., 2006).
- Life satisfaction. Satisfaction With Life Scale (SWLS, 5 items) (Diener et al., 1985).
- Self-esteem. Rosenberg Self-Esteem Scale (RSES, 10 items) (Rosenberg, 1965).
- Anaemia at 34 weeks. Recorded as yes or no.
- Haemoglobin at 34 weeks. Taken from the hospital laboratory report or the antenatal card.
- Parity. Recorded as: First pregnancy, Second or later.
- Age. Taken from the hospital laboratory report or the antenatal card.
- Residence. Recorded as: Urban, Rural.
- Procedure and ethics. The Janani cohort is simulated: every participant and every value was generated by computer to test Appuscholar, and no real person took part.
- Analysis. The main hypothesis was tested with an independent-samples t-test.

Results

- Results for each hypothesis follow.
- An independent-samples Welch t-test showed that GAD-7 score differed significantly between women with anaemia (M = 7.26, SD = 5.12) and women without anaemia (M = 5.33, SD = 4.24), t(986.89) = 7.03, p < .001, g = 0.42, 95% CI of the difference [1.39, 2.47].
- There was a moderate positive correlation between self-esteem (RSES) and life satisfaction (SWLS), which was statistically significant, r(1238) = .45, p < .001, 95% CI [.40, .49].
- There was a moderate negative correlation between GAD-7 score and life satisfaction (SWLS), which was statistically significant, r(1238) = -.35, p < .001, 95% CI [-.39, -.30].
- A paired-samples t-test showed a significant difference between haemoglobin at 16 weeks and haemoglobin at 34 weeks (M diff = -0.39, SD = 0.64), t(1239) = -21.50, p < .001, 95% CI [-0.43, -0.36], d = -0.61.
- A multiple linear regression with age, haemoglobin at 34 weeks, parity: First pregnancy (vs Second or later) as predictors of GAD-7 score was significant, F(3, 1236) = 11.24, p < .001, R^2 = .03, adjusted R^2 = .02. Individual predictors: age (β = .04, p = .120); haemoglobin at 34 weeks (β = -.16, p < .001); parity: First pregnancy (vs Second or later) (b = 0.23, p = .393).
- Internal consistency was α = .89, 95% CI [.88, .90], and McDonald's ω = .89, 95% CI [.88, .90] (k = 7 items, N = 1240).
- A chi-square test of independence showed no significant association between anaemia at 34 weeks and parity, χ^2(1, N = 1240) = 3.29, p = .070, Cramér's V = 0.05.
- A paired-samples t-test showed a significant difference between the mothers' GAD-7 scores and the fathers' GAD-7 scores (M diff = 1.58, SD = 4.82), t(611) = 8.12, p < .001, 95% CI [1.20, 1.97], d = 0.33.
- Distribution of GAD-7 score by anaemia at 34 weeks
- Note. Boxes show the median and interquartile range. Whiskers reach the most extreme values within 1.5 times the interquartile range. Each point is one participant.



Discussion

- In this simulated cohort, women who were anaemic at 34 weeks reported more anxiety than women who were not, with a small-to-moderate difference, so H1 was supported.
- Haemoglobin rose between 16 and 34 weeks, supporting H3 and in line with trials of daily iron in pregnancy, but four women in ten were still anaemic at 34 weeks.

Conclusion

- Anaemia and anxiety met in the same women in the third trimester.

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